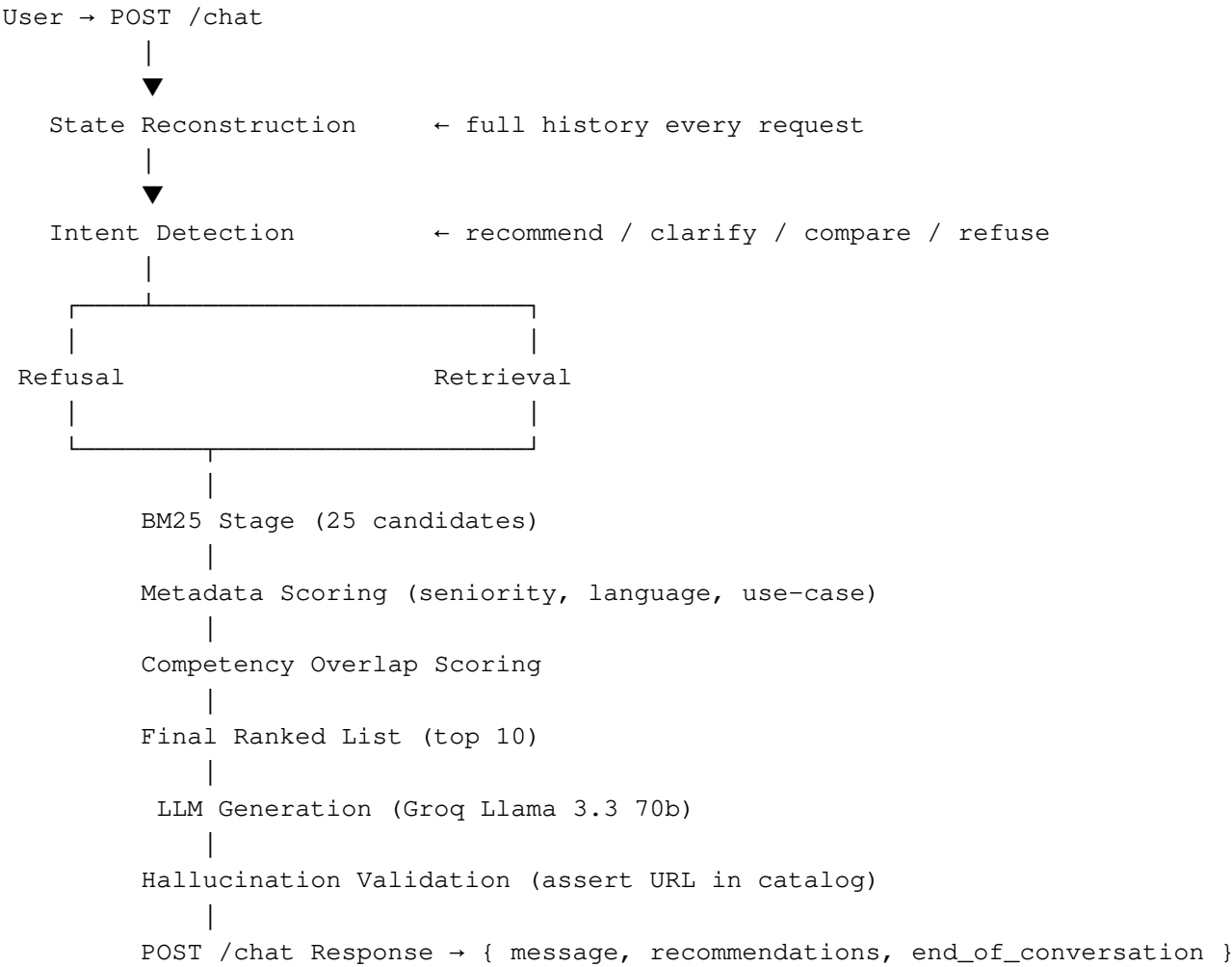


Approach Document

1. Architecture

The system is a conversational retrieval agent with a strict separation between retrieval (deterministic) and generation (LLM-based).



2. Retrieval Design

Why hybrid retrieval maximises Recall@10:

Stage 1 — BM25: Okapi BM25 with $k_1 = 1.5$, $b = 0.75$ over a composite document field (name + description + competencies + technical_domains + use_cases). Catches exact keyword signals like "Java", "Docker", "HIPAA", "DSI".

Stage 2 — Metadata Scoring: Five orthogonal signals combined with learned weights:

Signal	Weight	Rationale
BM25 (normalised)	0.35	Keyword recall anchor
Competency overlap	0.25	Role-skill alignment
Test type preference	0.15	User expressed needs
Seniority match	0.10	Avoids level mismatch
Use case match	0.10	Selection vs development

Tried: Per-question clarification — Burns turns. Switched to compound single-question policy.

Tradeoff: No persistent vector store — For a 60-item catalog, BM25 in-memory is sufficient and instant. At 1000+ items, would add FAISS or pgvector.

Tradeoff: Groq Llama 3.3 70b — High-speed API inference, large 128k context window, excellent markdown table instruction compliance.